

Laboratory of Computational Fluid Dynamics - 2025/2026

Exercise 2

Numerical Integration of the Linear Convection–Diffusion Equation

Gianmarco Bordin (ID 2157600)

Silvia Gallina (ID 2163528)

Matteo Marcandoro (ID 2163422)

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1 Introduction

The linear convection–diffusion equation is a fundamental partial differential equation (PDE) which retains the two basic physical mechanisms encountered in fluid mechanics, namely advection and viscous diffusion. The problem under consideration is:

$$\frac{\partial \phi(x, t)}{\partial t} = -c \frac{\partial \phi(x, t)}{\partial x} + \nu \frac{\partial^2 \phi(x, t)}{\partial x^2}, \quad x \in [0, L], \quad t \in [0, T] \quad (1)$$

where $c > 0$ is the constant convection velocity and $\nu \geq 0$ the constant diffusion coefficient. The boundary conditions are taken as periodic, $\phi(0, t) = \phi(L, t)$ for all $t \in [0, T]$, and we set $L = 1$. The integration time T as well as c and ν vary with the case considered.

The problem is integrated in time with a three-stage Runge–Kutta scheme (RK3), which is defined by a series of intermediate sub-stages followed by a linear combination producing the solution at the next time step. Therefore, for the three-stage version adopted here, we have:

$$\begin{aligned} \phi_1 &= \phi_n, \\ \phi_2 &= \phi_n + \alpha_2 \Delta t f(\phi_1), \\ \phi_3 &= \phi_n + \alpha_3 \Delta t f(\phi_2), \\ \phi_{n+1} &= \phi_n + \Delta t [\beta_1 f(\phi_1) + \beta_2 f(\phi_2) + \beta_3 f(\phi_3)]. \end{aligned} \quad (2)$$

The coefficients are constrained by the consistency requirement $\sum_k \beta_k = 1$ and by the order conditions imposed to maximize the formal accuracy. The implementation employed in this work uses the low-storage choice

$$\alpha_2 = \frac{1}{3}, \quad \alpha_3 = \frac{2}{3}, \quad \beta_1 = \frac{1}{4}, \quad \beta_2 = 0, \quad \beta_3 = \frac{3}{4}. \quad (3)$$

The resulting scheme is third-order accurate in time and its marginal stability curve, obtained from the amplification factor

$$A = \sum_{k=0}^{N_s} \frac{(\lambda \Delta t)^k}{k!}, \quad (4)$$

extends along the imaginary axis up to $|\lambda \Delta t| \approx \sqrt{3}$ and along the negative real axis up to $|\lambda \Delta t| \approx 2.5$. This makes RK3 well suited to the present problem since, as discussed below, the eigenvalues associated with the convective term are purely imaginary while those associated with the diffusive term are real and negative.

The first-order spatial derivative associated with the convective term is discretized with three different finite-difference stencils: first-order forward (FW1), first-order backward (BW1), and second-order centered (CS2):

$$\left. \frac{\partial \phi}{\partial x} \right|_{x_i}^{\text{FW1}} = \frac{\phi_{i+1} - \phi_i}{\Delta x}, \quad \left. \frac{\partial \phi}{\partial x} \right|_{x_i}^{\text{BW1}} = \frac{\phi_i - \phi_{i-1}}{\Delta x}, \quad \left. \frac{\partial \phi}{\partial x} \right|_{x_i}^{\text{CS2}} = \frac{\phi_{i+1} - \phi_{i-1}}{2 \Delta x}. \quad (5)$$

The spatial domain is discretized using a uniform grid of N_x points, defined as $x_i = i \Delta x$ for $i \in \{0, \dots, N_x - 1\}$, with a grid spacing of $\Delta x = L/N_x$. To enforce periodic boundary conditions,

we employ ghost nodes such that $\phi_{N_x} = \phi_0$ and $\phi_{-1} = \phi_{N_x-1}$.

The second-order spatial derivative associated with the diffusive term is always discretized with a fourth-order centered scheme (CS4):

$$\left. \frac{\partial^2 \phi}{\partial x^2} \right|_{x_i}^{\text{CS4}} = \frac{1}{\Delta x^2} \left[-\frac{1}{12} \phi_{i-2} + \frac{4}{3} \phi_{i-1} - \frac{5}{2} \phi_i + \frac{4}{3} \phi_{i+1} - \frac{1}{12} \phi_{i+2} \right]. \quad (6)$$

Four cases are addressed:

1. Pure convection with
 - 1.A. a Gaussian initial condition discretized via CS2;
 - 1.B. a harmonic initial condition discretized via FW1, BW1 and CS2;
2. Pure diffusion with a Gaussian initial condition discretized via CS4;
3. Full convection–diffusion with a Gaussian initial condition discretized via CS2 and CS4 for the convection and the diffusion term, respectively.

For each case, the CFL and diffusion numbers

$$\text{CFL} = \left| \frac{c \Delta t}{\Delta x} \right|, \quad D = \left| \frac{4 \nu \Delta t}{\Delta x^2} \right| \quad (7)$$

are evaluated.

For the analytical solutions, in the Gaussian case the exact solution

$$\phi(x, t) = \frac{\sigma}{\sqrt{\sigma^2 + 2\nu t}} \exp \left[-\frac{(x - x_0 - ct)^2}{2(\sigma^2 + 2\nu t)} \right] \quad (8)$$

is enforced to be periodic by summing a finite number of mirror images ($n = -3, \dots, 3$).

The harmonic case admits the closed-form periodic solution

$$\phi(x, t) = \exp \left[-\nu \left(\frac{2\pi}{L} \right)^2 t \right] \sin \left[\frac{2\pi}{L} (x - ct) \right]. \quad (9)$$

2 Numerical Implementation

2.1 Implementation A - Gianmarco Bordin

The numerical solver is developed in Modern Fortran, adopting a modular architecture to ensure flexibility and computational efficiency. This organization allows for a clear separation between the definition of physical parameters, spatial operators, and the core simulation engine.

2.1.1 Code Organization

The software is structured into several specialized modules, following a strict separation of concerns:

- **mod_param**: This module acts as a global repository for physical and numerical constants. It defines the floating-point precision (**rp** as **double precision**) and hosts variables such as the grid size (N_x), velocity (c), and viscosity (ν).
- **mod_spatial**: This module contains the implementation of the spatial derivative operators. It includes 1st-order schemes (FW1, BW1), 2nd-order centered schemes (CS2), and the 4th-order centered scheme (CS4) used for the diffusive term.
- **mod_rk3**: This module isolates the time integration algorithm. It implements the explicit 3rd-order Runge-Kutta (RK3) method, effectively handling the intermediate fractional time steps required to advance the solution while ensuring high temporal accuracy and stability.
- **mod_exact**: This module provides the analytical solutions for the assigned test cases (e.g., pure convection of a harmonic wave or diffusion of a Gaussian pulse). It computes the theoretical benchmarks evaluated at specific spatial coordinates and time instants, allowing for rigorous code validation. For the Gaussian, a line of code ensures the periodic wrapping.
- **mod_simulator**: The core engine of the program. It contains the **run_simulation** subroutine, which acts as a bridge between the physical parameters, the spatial operators, and the time integrator, effectively managing the simulation loop and memory allocation.
- **main.f90**: The entry point of the application. Following a task-oriented approach, the parameters are redefined at each task within the main program, ensuring full control over the specific simulation settings (e.g., CFL and Fourier numbers).

2.1.2 Algorithmic Workflow

The simulation execution follows a rigorous sequential logic:

1. **Initialization and Grid Generation**: The domain L is discretized using a cell-centered grid approach. The cell centers are defined as $x_i = (i - 0.5)\Delta x$, ensuring that periodic boundary conditions are naturally handled without node overlapping at the boundaries.
2. **Memory Management**: Dynamic memory allocation is used for the state arrays ϕ and x . This allows the solver to handle different grid resolutions (N_x) within the same execution without recompilation.
3. **Initial Condition (IC)**: The field is initialized based on the task requirements, using either a Gaussian pulse or a harmonic wave.
4. **Time Integration**: The solution is advanced in time using an explicit 3rd-order Runge-Kutta (RK3) scheme. At each sub-step, the Right-Hand Side (RHS) of the equation is

computed as:

$$\text{RHS} = -c \frac{\partial \phi}{\partial x} + \nu \frac{\partial^2 \phi}{\partial x^2} \quad (10)$$

5. **Spatial Discretization Logic:** To optimize performance, the code evaluates the convective term only if $c \neq 0$ and the diffusive term only if $\nu > 0$. The convective scheme is selected via a `select case` construct, while the diffusive term consistently employs the CS4 operator.
6. **Data Output:** At specified intervals, the numerical solution and the analytical solution are exported to text files for post-processing, which has been done using Matlab.

2.1.3 Stability and Safety Features

To ensure the robustness of the solver, two main safety mechanisms were implemented:

- **Adaptive Logic:** The convection scheme selector defaults to a termination signal if an unknown scheme is requested, preventing the propagation of numerical errors.
- **Divergence Control:** A safety threshold is monitored during the time loop. If the maximum value of the solution exceeds a critical limit (e.g., 10^6), the simulation is aborted to prevent floating-point exceptions (IEEE overflows) and corrupted data output.
- **CFL and D:** Security warnings are printed in the terminal if CFL and D numbers have a value bigger than 1.

2.2 Implementation B - Silvia Gallina

The code is implemented in Fortran 90 and organized into four modules plus one driver program per task.

The module `mod_param` defines the working precision (double precision) and the constant π . The module `mod_schemes` collects the spatial discretizations: a second-order centered (CS2), a first-order forward (FW1) and a first-order backward (BW1) scheme for the convective term, and a fourth-order centered scheme (CS4) for the diffusive term, all with periodic boundary conditions enforced through index wrapping.

A wrapper subroutine `computerhs` selects the desired schemes at run time and assembles the total right-hand side as the sum of the convective and diffusive contributions; setting either flag to "none" disables the corresponding term, so that the same routine can be used for pure convection, pure diffusion or the full convection-diffusion problem.

Time integration is carried out in `mod_rk3` through the three-stage third-order Runge–Kutta scheme that calls `computerhs` at each stage.

Finally, `mod_save` writes the grid, the numerical and the analytical solutions to a text file at selected time steps, duplicating the last point at $x = L$ to ease plotting on the periodic domain.

For each task, a separate program sets the physical and numerical parameters, defines the initial condition, chooses the schemes, and then advances the solution in time by calling `rk3` at every step, saving the results at regular intervals. In all cases the CFL number and/or the diffusion number are printed at the start of the simulation to verify that the stability limits of the explicit scheme are respected.

The most significant code snippets are provided in Appendix ??.

2.3 Implementation C - Matteo Marcandoro

The numerical code is written in Fortran 90 in double precision through the intrinsic function `selected_real_kind(15, 307)`. Post-processing is performed via Python scripts based on `matplotlib`, `pandas` and `numpy`. The code rely on a module `ex2_module` that compute:

1. the initial-condition functions (Gaussian and harmonic),
2. the corresponding analytical solutions used for comparison,
3. three finite-difference operators for the first derivative (FW1, BW1, CS2) applied to the convective term,
4. a fourth-order centred operator for the second derivative (CS4) applied to the diffusive term,
5. and three third-order Runge–Kutta subroutines (`rk3_FW1_CS4`, `rk3_BW1_CS4`, `rk3_CS2_CS4`), each combining one of the convective operators with CS4.

Periodic boundary conditions are enforced inside every finite-difference routine through ghost nodes copied from the opposite end of the array.

Four main programs (`ex_2_1A`, `ex_2_1B`, `ex_2_2`, `ex_2_3`) cover the four test cases. Each program sets the physical and numerical parameters, defines the uniform spatial grid $x_i = i \Delta x$ with $\Delta x = L/N_s$ (so that $x_{N_s} \equiv x_0$ by periodicity), computes the stability numbers CFL and D , allocates the numerical and exact solution arrays, and advances the solution in time by calling the appropriate RK3 subroutine.

Both numerical and exact fields are saved to text files in a tabular format. The first row is the header and contains the spatial grid coordinates. Then, at each time step, a new row is added below containing the current time value followed by the solution vector over the spatial grid. Finally, the last row contains the corresponding stability parameters CFL and D .

In the python scripts the output text files are imported as `numpy` arrays and converted into `pandas` DataFrames for easy data handling. Finally, the solution curves at selected time steps are plotted using `matplotlib`.

The complete source code is reported in Appendix ??.

3 Results

3.1 Task 1A - Pure convection with Gaussian initial condition (CS2)

We consider the pure convection problem ($\nu = 0$, $c = 0.1$) with a Gaussian initial condition

$$\phi(x, 0) = \exp\left[-\frac{(x - x_0)^2}{2\sigma^2}\right], \quad (11)$$

centered at $x_0 = L/2$ with standard deviation $\sigma = 0.1$. The integration is performed over $t \in [0, 1]$ with $\Delta t = 10^{-2}$ and $N_x = 100$ grid points, which yields $\Delta x = 10^{-2}$ and a CFL number

$$\text{CFL} = \left| \frac{c \Delta t}{\Delta x} \right| = \frac{0.1 \cdot 10^{-2}}{10^{-2}} = 0.1,$$

well inside the stability region of the RK3 scheme. The first-order derivative is computed with the second-order centered (CS2) scheme.

Figure 1 shows the numerical solution and the exact solution at five equally spaced time instants. The Gaussian profile is convected from left to right at velocity c without any appreciable change in shape or amplitude, in agreement with the analytical solution. This is consistent with the fact that the Gaussian profile is dominated by low-wavenumber components: since the modified wavenumber of CS2 is virtually indistinguishable from k_n at low $k_n \Delta x$, the dispersion error remains negligible over the entire integration interval.

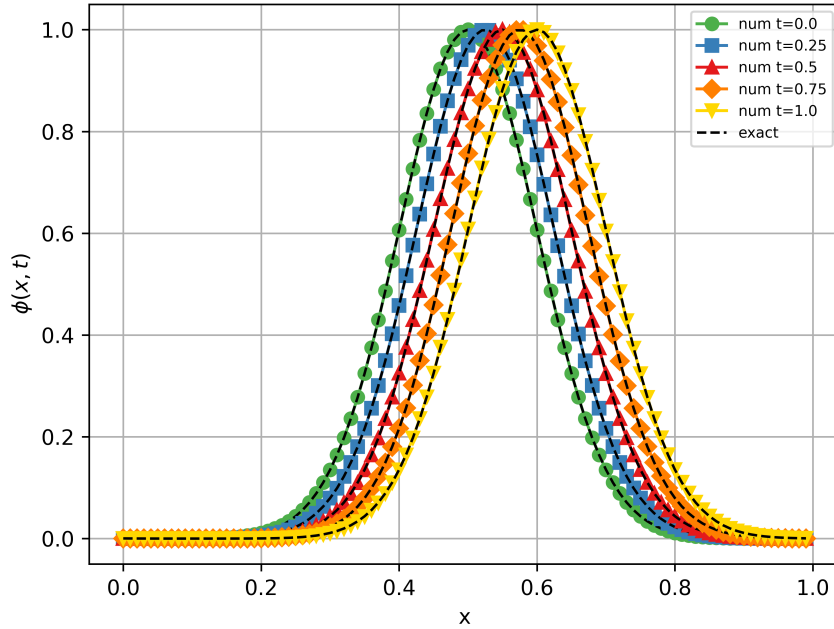


Figure 1: Numerical (CS2) and exact solution of the pure convection problem with Gaussian initial condition at five equally spaced time instants (CFL = 0.1).

3.2 Task 1B - Pure convection with harmonic initial condition (FW1, BW1, CS2)

We consider the pure convection problem ($\nu = 0$, $c = 0.1$) with a harmonic initial condition

$$\phi(x, 0) = \sin\left(\frac{2\pi}{L}x\right). \quad (12)$$

The integration is performed over $t \in [0, 100]$ with $\Delta t = 0.25$ and $N_x = 50$ grid points, which yields $\Delta x = 0.02$ and

$$\text{CFL} = \left| \frac{c \Delta t}{\Delta x} \right| = \frac{0.1 \cdot 0.25}{0.02} = 1.25.$$

This value still lies inside the stability region of the RK3 scheme along the imaginary axis ($\sqrt{3} \approx 1.73$), so that no time-integration instability is expected for the centered scheme. The three first-order discretizations (FW1, BW1, CS2) are tested separately and the corresponding solutions are presented in Figures 2, 3, 4, 5 and Figure 6.

3.2.1 First-order forward scheme (FW1)

The forward scheme (FW1, Figure 2) blows up after a handful of time steps. This is the expected behaviour. For the FW1 scheme, the imaginary part of the modified wavenumber satisfies

$$\Im(k'_m) \geq 0 \text{ for all } m,$$

combined with $c > 0$, this yields a positive growth rate $c \Im(k'_m) \geq 0$ for every Fourier mode, leading to an unconditional artificial amplification. The scheme is therefore unconditionally unstable in this configuration, regardless of the value of CFL. Note that, in order to keep the figure readable, only the first 20 time steps are displayed in the plot.

3.2.2 First-order backward scheme (BW1)

The backward scheme (BW1, Figure 3) is on the contrary stable, in agreement with the fact that for $c > 0$ the imaginary part of its modified wavenumber is non-positive and produces a decay of the Fourier modes. The price for this stabilizing effect is a substantial artificial dissipation: the harmonic signal is heavily damped after only a few periods, and its amplitude is essentially zero already at $t \sim 50$. The dispersion error of BW1, identical to that of CS2, is largely masked by the dominant dissipation. Note that, here, only the first 500 time steps are shown ($t \leq 125$), since beyond that the solution is indistinguishable from zero.

3.2.3 Second-order centered scheme (CS2)

The centered scheme (CS2, Figure 4, Figure 5 and Figure 6) is purely dispersive: the modified wavenumber is real, so no artificial dissipation is introduced and the amplitude of the solution is preserved at all times. However, the phase velocity is wavenumber-dependent and, for the rather coarse spatial resolution employed here ($k_1 \Delta x = 2\pi/N_x \approx 0.13$), a non-negligible phase error accumulates over the long time horizon. As a consequence, the numerical solution progressively drifts out of phase with respect to the exact one, while keeping the correct amplitude (as shown in Figure 6). This behaviour is especially clear at the largest displayed time instants (Figure 5), where the numerical curve is visibly shifted with respect to the analytical solution.

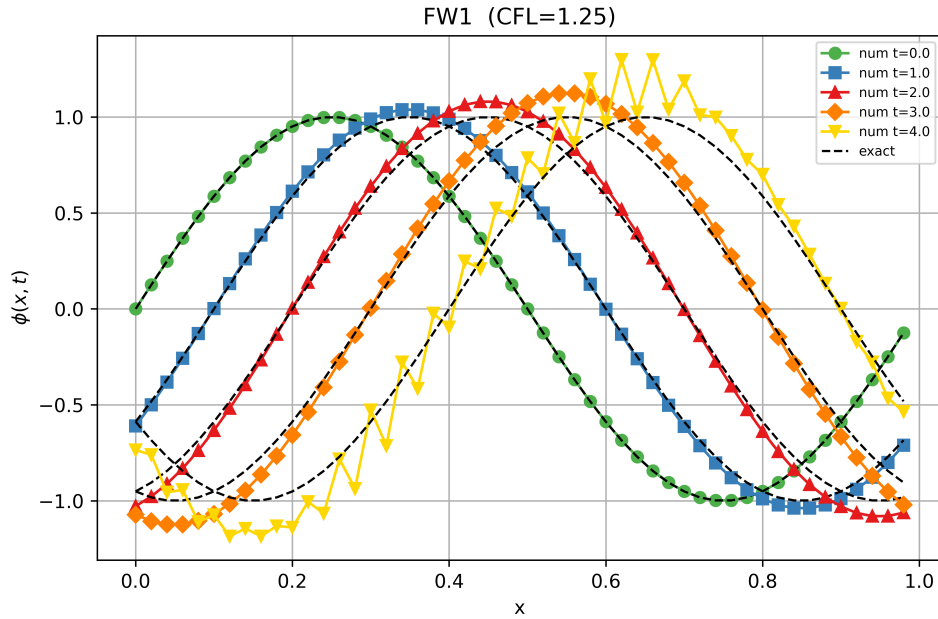


Figure 2: Numerical (FW1) and exact solution of the pure convection problem with harmonic initial condition ($CFL = 1.25$). Only the first 20 time steps are displayed: the scheme is unconditionally unstable for $c > 0$.

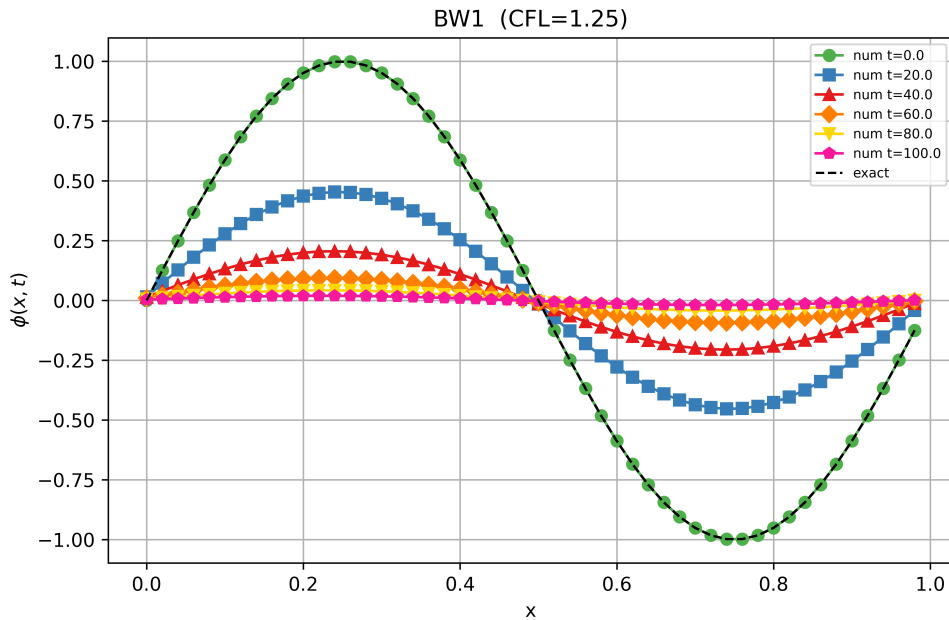


Figure 3: Numerical (BW1) and exact solution of the pure convection problem with harmonic initial condition ($CFL = 1.25$). The scheme is stable but introduces a strong artificial dissipation that quickly damps the signal.

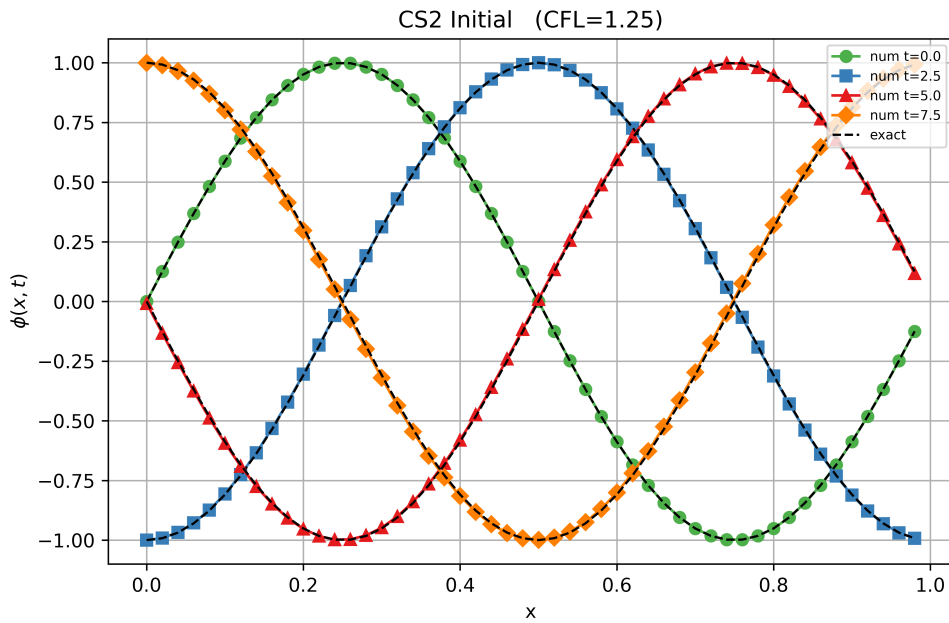


Figure 4: Numerical (CS2) and exact solution of the pure convection problem with harmonic initial condition (CFL = 1.25) at initial time steps.

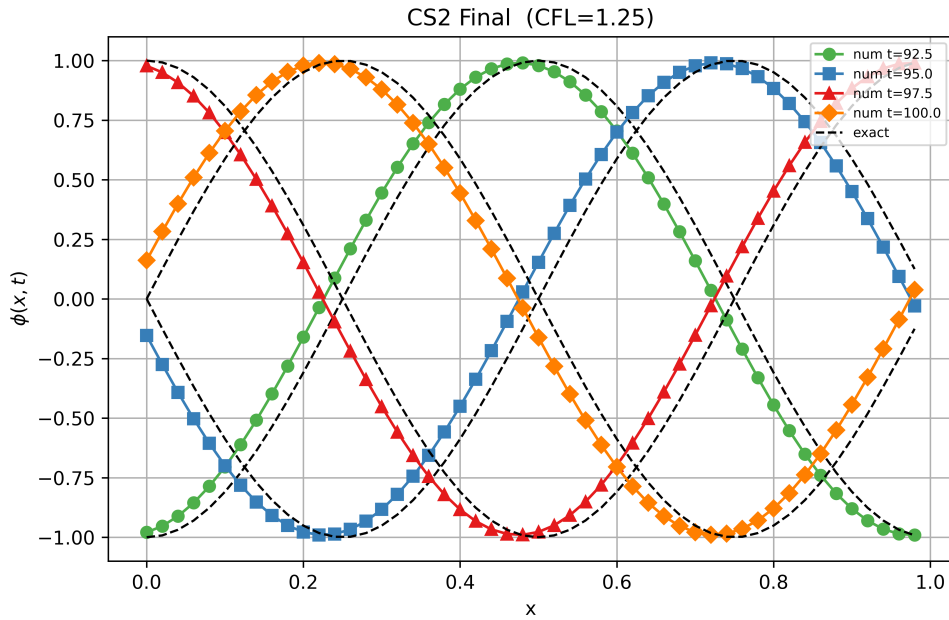


Figure 5: Numerical (CS2) and exact solution of the pure convection problem with harmonic initial condition (CFL = 1.25) at final time steps.

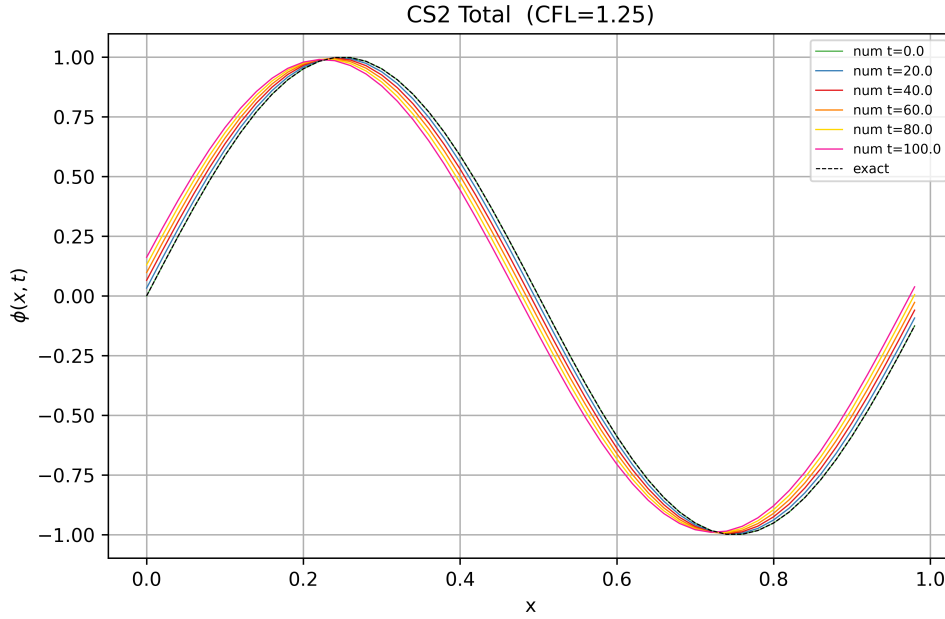


Figure 6: Numerical (CS2) and exact solution of the pure convection problem with harmonic initial condition (CFL = 1.25) at periodic time instants.

3.3 Task 2 - Pure diffusion, Gaussian initial condition (CS4)

We consider the pure diffusion problem ($c = 0$, $\nu = 0.01$) with a Gaussian initial condition centered at $x_0 = L/2$ and standard deviation $\sigma = 0.1$. The integration is performed over $t \in [0, 3]$ with $\Delta t = 10^{-3}$ and $N_x = 100$ grid points, which yields $\Delta x = 10^{-2}$ and a diffusion number

$$D = \left| \frac{4\nu\Delta t}{\Delta x^2} \right| = \frac{4 \cdot 0.01 \cdot 10^{-3}}{(10^{-2})^2} = 0.4,$$

well inside the stability region of the RK3 scheme along the negative real axis.

Figure 7 shows the numerical solution obtained with the fourth-order centered (CS4) stencil at five equally spaced time instants together with the exact solution. The Gaussian peak is progressively flattened and broadened as time advances, in excellent agreement with the analytical solution. The CS4 stencil reproduces accurately the second-derivative operator over the entire wavenumber range relevant for this problem: since the Gaussian initial condition is dominated by low wavenumbers, the modified wavenumber $(k_n^2)_m^{CS4}$ is essentially indistinguishable from k_n^2 and no significant numerical artifact is introduced.

3.4 Task 3 - Convection-diffusion with Gaussian initial condition (CS2 + CS4)

We finally consider the full convection–diffusion problem with $c = 0.2$ and $\nu = 0.01$, again with a Gaussian initial condition centered at $x_0 = L/2$ and standard deviation $\sigma = 0.1$. The integration is performed over $t \in [0, 5]$ with $\Delta t = 10^{-3}$ and $N_x = 100$ grid points, yielding $\Delta x = 10^{-2}$ and

$$\text{CFL} = \frac{0.2 \cdot 10^{-3}}{10^{-2}} = 0.02, \quad D = \frac{4 \cdot 0.01 \cdot 10^{-3}}{(10^{-2})^2} = 0.4.$$

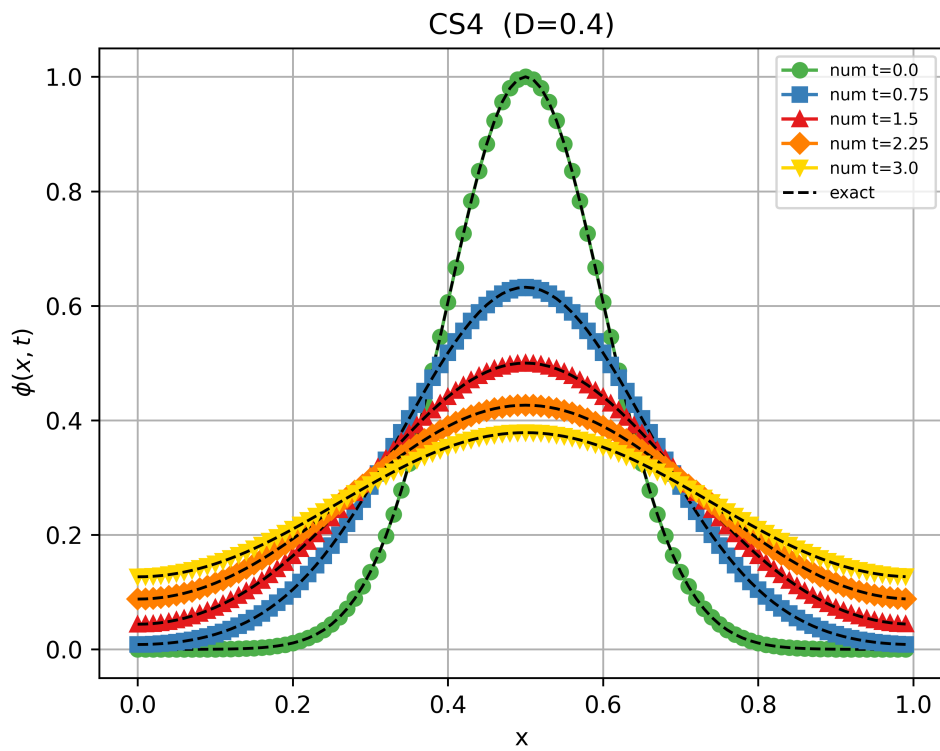


Figure 7: Numerical (CS4) and exact solution of the pure diffusion problem with Gaussian initial condition at five equally spaced time instants. $D = 0.4$.

Both numbers lie comfortably inside the RK3 stability region. The convective term is discretized with the second-order centered (CS2) stencil and the diffusive term with the fourth-order centered (CS4) stencil. This combination is a natural choice for the present problem: CS2 is non-dissipative and matches the spectral behaviour of the smooth Gaussian profile without contaminating it with artificial damping, while CS4 provides a high-order representation of the second derivative that accurately reproduces the diffusive smoothing.

Figure 8 shows the numerical and exact solutions at five equally spaced time instants. The Gaussian profile is simultaneously transported at velocity c and progressively flattened and broadened by diffusion. The numerical solution follows the analytical one very closely throughout the integration interval, both in terms of position of the peak and of amplitude/width, demonstrating that the chosen scheme combination correctly captures the interaction between advection and diffusion.

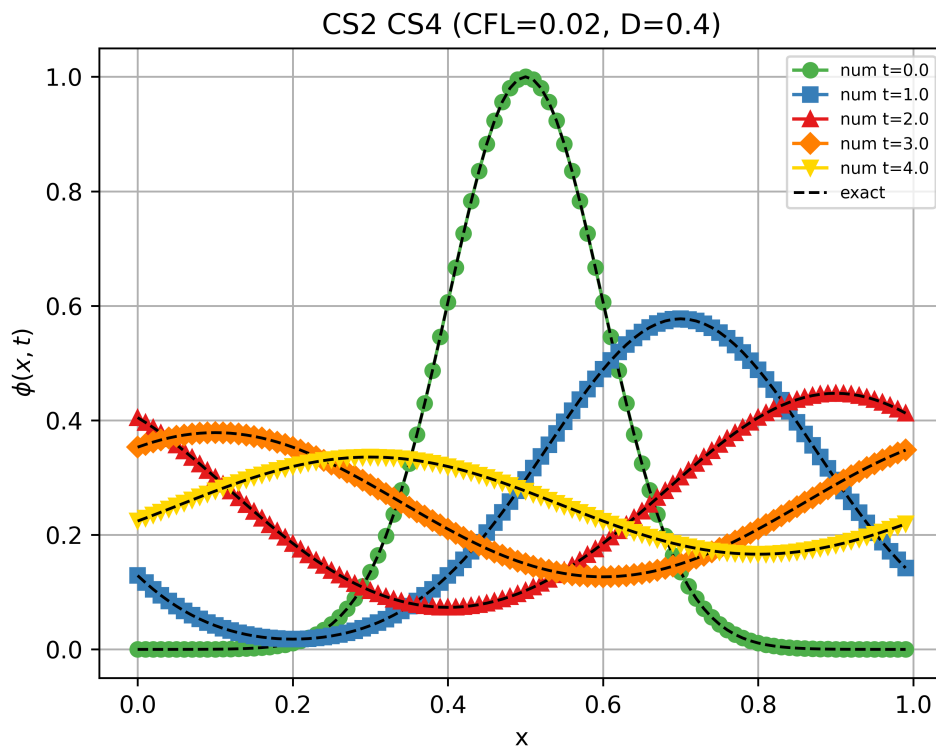


Figure 8: Task 3. Numerical (CS2 for the convective term, CS4 for the diffusive term) and exact solution of the convection–diffusion problem with Gaussian initial condition. $CFL = 0.02$, $D = 0.4$.

4 Conclusions

The numerical experiments confirm the theoretical predictions of the spectral analysis of finite-difference schemes for the convection-diffusion equation. For pure convection with smooth, low-wavenumber initial data (Task 1A) the centered CS2 scheme provides a faithful representation of the propagation. The harmonic test of Task 1B unambiguously highlights the role played by the imaginary part of the modified wavenumber: FW1 amplifies the Fourier modes and is unconditionally unstable for $c > 0$, BW1 is stable but heavily dissipative, and CS2 preserves

the amplitude at the price of a progressive phase shift related to its dispersion error. The pure diffusion problem (Task 2) is accurately solved by the high-order CS4 stencil. Finally, the combined convection-diffusion problem (Task 3) is well captured by the CS2+CS4 combination, which correctly reproduces both the transport of the Gaussian peak and its diffusive broadening. In all cases, the choice of Δt and Δx keeps the corresponding CFL and D numbers within the stability region of the RK3 time integration.